

# 1. What is Salesforce?

Salesforce is the world's leading Cloud-based CRM (Customer Relationship Management) platform. It helps businesses store, manage, and analyze customer data and provides tools to automate sales, service, marketing, analytics, and integrations—all in the cloud, without heavy IT setup.

👉 In short: Salesforce helps companies connect with customers, streamline business processes, and grow faster.

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## Why Should We Use Salesforce?

### 1. Cloud-Based & Accessible Anywhere

- No need to install software or maintain servers.
- Access Salesforce from any device (laptop, mobile, tablet) with just an internet connection.

### 2. Customization & Flexibility

- Highly customizable through clicks (point-and-click tools) and code (Apex, LWC, APIs).
- Can adapt to almost any business model (Sales, Service, Marketing, Finance, Education, Healthcare, etc.).

### 3. Scalability

- Works for small startups to Fortune 500 companies.
- Scales with your business growth.

### 4. AppExchange Marketplace

- Thousands of prebuilt apps (like App Store/Play Store) that can be directly installed to extend functionality.

### 5. Automation

- Automates repetitive tasks with Flows, Process Builder, Approvals, OmniStudio, Einstein AI.
- Saves time and reduces errors.

### 6. Integration Capabilities

- Seamlessly integrates with ERP systems, social media, email, analytics, and third-party applications via REST/SOAP APIs and MuleSoft.

### 7. Security

- Robust data security model: Profiles, Roles, Permission Sets, Sharing Rules, OWD (Organization-Wide Defaults).
- Meets global compliance standards (GDPR, HIPAA, etc.).

### 8. Analytics & Reporting

- Real-time dashboards and reports for 360° visibility of business performance.
- Helps in data-driven decision-making.

### 9. Continuous Innovation

- Salesforce releases 3 major updates each year (Spring, Summer, Winter), always adding new features without extra cost.

#### 10. Strong Ecosystem

- Massive community, training resources (Trailhead), certified professionals, and career opportunities
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## 2. Why Sandboxes are Used?

1. **Safe Development** – Build and test new features without breaking live data.
  2. **Testing** – Run QA, UAT (User Acceptance Testing), and performance testing.
  3. **Training** – Train new users on real-like data without exposing production.
  4. **Deployment** – Validate deployments before moving to production.
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### Types of Sandboxes in Salesforce

#### 1. Developer Sandbox

- Copy of metadata only (no production data).
  - Small storage (200 MB data).
  - Best for individual development and unit testing.
  - Refresh interval: once per day.
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#### 2. Developer Pro Sandbox

- Similar to Developer, but with more storage (1 GB).
  - Can hold larger data sets for testing.
  - Good for development + integration testing.
  - Refresh interval: once per day.
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#### 3. Partial Copy Sandbox

- Includes metadata + sample data (subset of production data defined by a Sandbox Template).
  - Storage: 5 GB of data, 10,000 records per selected object.
  - Useful for testing with real-like data.
  - Refresh interval: every 5 days.
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#### 4. Full Sandbox

- Complete copy of metadata + all production data.

- **Storage** = size of production org.
- **Ideal for performance testing, load testing, and UAT.**
- **Refresh interval: every 29 days.**
- **Supports sandbox seeding for masking data (to secure sensitive info)**

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### 3. What is a Profile in Salesforce?

A **Profile** in Salesforce is a collection of **settings and permissions** that defines what a user can do in Salesforce. It acts as the **blueprint of a user's access** to objects, records, fields, tabs, apps, and system functionality.

Think of it as:

👉 *"Profile = What a user can **see** and **do** in Salesforce."*

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#### ◆ Key Responsibilities of a Profile

1. **Object Permissions** – Create, Read, Edit, Delete (CRUD) on Standard & Custom objects.
2. **Field-Level Security** – Visibility and editability of individual fields.
3. **Tab Settings** – Default On, Default Off, or Hidden.
4. **App Settings** – Which apps are visible to the user.
5. **Record Types & Page Layouts** – Assign record types and layouts to users.
6. **System Permissions** – Access to functionalities like API Enabled, View All Data, Modify All Data, Password policies, etc.
7. **Login Settings** – Login hours and Login IP ranges restrictions.
8. **Administrative Permissions** – Example: "Manage Users", "Customize Application".

⚡ Each user in Salesforce **must be assigned exactly one Profile**.

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### 4. What is a Role in Salesforce?

A **Role** in Salesforce is part of the **Role Hierarchy**, which controls **record-level access** (who can see what records).

- Roles are assigned to users.
- They define where a user stands in the data visibility hierarchy.
- Users higher up in the hierarchy can automatically **see, edit, and report** on all the data owned by or shared with their subordinates.

👉 Example:

- CEO → Sales Director → Sales Manager → Sales Rep

- If a Sales Rep owns an Opportunity, the Sales Manager, Director, and CEO (above them in hierarchy) can also see it.

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#### ◆ Key Points about Roles

- Roles are **optional**, but usually implemented in orgs where record visibility needs to be controlled.
- Used mainly for **sharing** purposes (not for field-level access).
- Roles affect **Reports, Dashboards, and Record Visibility**.
- You can define “**Grant Access Using Hierarchies**” for standard & custom objects (on object’s sharing settings).

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## 5. What is a user in Salesforce?

In Salesforce, a **User** represents an individual who logs in and interacts with the Salesforce platform. Every person who needs access to Salesforce data and applications must have a User record.

A **User record** defines the following:

- **Identity** of the person (Name, Email, Username, Alias).
- **Login access** (Username, Password, Role, Profile, Permission Sets).
- **Licenses** (which determine available features).
- **Settings** (Locale, Time Zone, Language).

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## 6. Permission Set in Salesforce

### Definition

A **Permission Set** in Salesforce is a collection of **settings and permissions** that give users access to additional features or functions **without changing their Profile**.

- Think of it as an **add-on layer** on top of a user’s profile.
- Profiles define the **baseline level of access** (minimum permissions).
- Permission sets give **extra access temporarily or permanently**, without needing to clone or modify the profile.

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### Why Permission Sets are Used?

1. **Avoid Profile Proliferation**  
Instead of creating multiple profiles for small differences in access, you can use a single profile and add permission sets where needed.
2. **Granular Access Control**  
You can give users **extra permissions only when required** (e.g., one user needs API access, another needs Modify All on an object).

### 3. Flexibility

Easily assign/remove permissions **without editing profiles**.

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## 7. What is App, object and Fields in salesforce?

### 1. App in Salesforce

#### Definition

- An **App** in Salesforce is a collection of **tabs, objects, and functionalities** grouped together to serve a specific business process.
- Think of it as a **container** that holds everything related to a particular business function (like Sales, Service, or a custom HR app).

#### Types of Apps

1. **Standard Apps** – Prebuilt by Salesforce (Sales, Service, Marketing).
2. **Custom Apps** – Created by admins/developers for specific needs (e.g., HR Management).
3. **Connected Apps** – Used for external integrations (e.g., integrating Salesforce with Google Drive).

#### Limitations

- No strict limit on number of apps, but **too many apps may cause UI clutter**.
  - User license may restrict **which apps** a user can access.
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### 2. Object in Salesforce

#### Definition

- An **Object** is like a **database table** that stores records (rows).
- Each record is a set of **fields** (columns).
- Objects represent **business entities** (e.g., Account, Contact, Opportunity, Invoice).

#### Types of Objects

1. **Standard Objects** – Predefined by Salesforce (Account, Contact, Lead, Opportunity, Case).
2. **Custom Objects** – Created by users to capture specific business data (e.g., Property\_\_c, Invoice\_\_c).
3. **External Objects** – Used in **Salesforce Connect** to map data stored outside Salesforce.
4. **Big Objects** – Handle massive data volumes (billions of records).

#### Limitations

- Max **Custom Objects per org**: depends on edition (Enterprise: 200, Unlimited: 2000, etc.).
- **Big Objects** support limited features (e.g., no triggers on standard DML).
- Cannot exceed **total storage limits** of the org.

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### 3. Field in Salesforce

#### Definition

- A **Field** in Salesforce is like a **column in a database table**.
- It stores a piece of data for each record (e.g., Name, Email, Industry, Amount).

#### Types of Fields

- **Standard Fields** – Provided by Salesforce (Id, CreatedDate, Owner).
- **Custom Fields** – Defined by users (e.g., PAN\_Number\_\_c).
- **Formula Fields** – Calculated dynamically (e.g., Total = Quantity × Price).
- **Roll-up Summary Fields** – Summarize child records (only on Master-Detail relationships).
- **Lookup/Master-Detail Fields** – Relationship fields between objects.

#### Limitations

- **Max Custom Fields per object**: depends on edition (Enterprise: 500, Unlimited: 800).
  - **Formula Fields**: max 3,900 characters for formula expression.
  - **Roll-up Summary Fields**: max 25 per object (only on master side).
  - Cannot create Roll-up on **Lookup relationships** (requires code or tools).
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## 8. Salesforce Relationships

#### Definition

A **relationship in Salesforce** defines how two objects are **connected** and how data is shared between them.

#### Types of Relationships in Salesforce

There are mainly **two categories**:

1. **Object Relationships (Schema level)**
  2. **Record Relationships (Data level)**
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#### 1. Object Relationships

These are defined at the schema (object/field) level.

##### a) Lookup Relationship

- A **loose relationship** between two objects.
- Child record can exist **without a parent**.
- Maximum 40 lookup fields per object.
- Example: A Contact linked to an Account (but can exist without one).

👉 **Use case:** When relationship is optional.

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#### b) Master-Detail Relationship

- A **tight relationship** between two objects.
- Child record **cannot exist without a parent**.
- If the parent is deleted → child is also deleted (**cascade delete**).
- Child record inherits parent's **ownership, sharing, and security**.
- Each object can have **up to 2 master-detail relationships**.
- Example: An Expense Line Item (child) must be linked to an Expense Report (parent).

👉 **Use case:** When you want strict dependency.

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#### c) Many-to-Many Relationship

- Achieved using a **junction object** (custom object with two master-detail relationships).
- Allows linking **multiple records of one object to multiple records of another**.
- Example: A student can enroll in many courses, and a course can have many students.

👉 **Use case:** When both sides can have multiple relationships.

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#### d) External Lookup Relationship

- Links a child Salesforce object to a parent **external object** (via Salesforce Connect).
- Parent external object data lives outside Salesforce.

👉 **Use case:** Integrating with external databases

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### 2. Record Relationships

These are relationships between records (not schema-level).

#### e) Self-Relationship

- An object relates to itself via a lookup field.
- Example: An Employee record can have a "Manager" lookup to another Employee.

#### f) Hierarchical Relationship

- A special lookup relationship **only available on the User object**.
  - Allows defining "who reports to whom" within users.
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## 9. What is Field History Tracking in Salesforce?

- **Field History Tracking** in Salesforce is a feature that allows you to **track changes to the values of specific fields** on standard and custom objects.
- It records **old value, new value, the user who made the change, and the date/time of change**.
- Example: If you enable history tracking on the **Status** field of Case, whenever the Status changes from *Open* to *Closed*, Salesforce will automatically log this change in the history related list.

### ⚡ Key Points:

- You can track up to **20 fields per object** (standard/custom).
- Salesforce stores field history for **18 months in the org** and **up to 24 months via API**.
- Tracked history can be viewed in the **Field History related list** on the object's record page.

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## 10. What is a Page Layout in Salesforce?

- A **Page Layout** in Salesforce controls **how fields, related lists, sections, and buttons** are displayed on a record's detail page and edit page.
- It helps define the **user interface (UI)** for records of a particular object.
- In short → it decides **what a user can see, where it is placed, and whether it's read-only, required, or hidden**.

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### Key Features of Page Layouts

1. **Arrange Fields** → Decide the order of fields on the record page.
2. **Make Fields Required/Read-only** → At layout level (not at database level).
3. **Add Sections** → Group fields logically (e.g., *Contact Info, Billing Info*).
4. **Control Related Lists** → Choose which related lists (e.g., *Opportunities, Cases*) appear.
5. **Customize Buttons & Actions** → Decide which **buttons, links, and actions** show on a record.
6. **Assign Different Layouts to Profiles** → Different users (based on role/profile) can see different layouts for the same object.

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## 11. What is a Lightning Record Page in Salesforce?

- A **Lightning Record Page** is a **customized layout for a record page** in the **Lightning Experience**.
- It controls **where components (fields, related lists, charts, custom components, etc.) appear on the record page**.
- Built using the **Lightning App Builder**.
- Think of it as:
  - **Page Layout** → controls **fields, related lists, and buttons**.



- **Lightning Record Page** → controls the overall structure and components of the record page UI.

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### Key Features

1. **Custom Components** → You can place standard or custom Lightning components (like Highlights Panel, Path, Related List – Quick Links, Charts, etc.).
2. **Flexible Layouts** → Drag-and-drop interface for sections like header, tabs, sidebars, etc.
3. **Dynamic Pages** → Show or hide components based on **field values, user profile, or record criteria**.
4. **Multiple Assignments** → You can assign different Lightning Record Pages to:
  - App
  - Record Type
  - Profile

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## 12. Field Dependency

In Salesforce, **Field Dependency** is a feature that lets you control the visibility of picklist values based on the value selected in another picklist.

### Example:

- **Controlling Field:** Country
- **Dependent Field:** State

If a user selects **India** in the Country field, the State picklist will only show **Tamil Nadu, Karnataka, Telangana, etc.**

If a user selects **USA**, the State picklist will only show **California, Texas, New York, etc.**

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### Key Points about Field Dependencies:

1. **Controlling Field Types**
  - Standard or custom picklist (single-select)
  - Checkbox
2. **Dependent Field Types**
  - Picklist (single-select or multi-select)
3. **Limitations**
  - A controlling field can control up to **300 values**.
  - A field dependency can have up to **10 controlling fields per object**.
  - Standard multi-select picklists cannot be controlling fields.
4. **Where it is useful**
  - Data consistency (avoids irrelevant values).

- Improves user experience (shows only valid options).
- Reduces errors in data entry.

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## 13. What is a Field Set in Salesforce?

A **Field Set** is a grouping of fields on an object that you can define and then use in **Visualforce pages, Apex, or managed packages** to dynamically display or process fields.

Instead of hardcoding field names in code/UI, you can use a field set to make it **configurable** for admins.

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### ◆ Example:

Imagine you have a **Student\_\_c** object.

You create a **Field Set** called **Student\_Info** that contains:

- Name
- Age
- Class
- Parent Contact

Now, in a Visualforce page or LWC, instead of hardcoding those fields, you just reference **Student\_Info**.

If tomorrow the admin adds **Address** to the field set, it will automatically appear on the page without changing code.

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### ◆ Where Field Sets are used?

1. **Visualforce Pages** – display fields dynamically.
2. **Apex Code** – query only the fields included in the field set.
3. **Managed Packages** – make packages flexible by letting customers configure fields.
4. **Search Forms / Dynamic UI** – useful in custom apps to let admins control visible fields.

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## 14. What is a Compact Layout?

A **Compact Layout** in Salesforce controls which **key fields** are displayed in the **record highlights section** (the top of a record page in Lightning Experience and Salesforce Mobile App).

It gives users a **quick view of the most important information** without scrolling through the whole record.

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### ◆ Example

For **Contact** object:

- You create a compact layout that shows:
  - Name
  - Phone
  - Email
  - Account Name

When a user opens a Contact record in Lightning or Mobile, they immediately see these 4 fields at the top (record highlights panel).

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◆ **Where are Compact Layouts used?**

- **Lightning Experience** → Record highlights panel (top of the record page).
- **Salesforce Mobile App** → Record header section.
- **Console View** → Record preview cards.

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◆ **Key Points**

- You can add **up to 10 fields** in a compact layout.
- Only fields on the object (not related object fields like formula with cross-object reference) can be added.
- The **first field** in the compact layout is shown in bold as the record's main identifier.
- Each object can have **multiple compact layouts**, but only one can be assigned as the **primary default** for profiles via a **Record Type → Compact Layout Assignment**.

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## 15. What is a Record Type in Salesforce?

A **Record Type** in Salesforce allows you to **offer different business processes, picklist values, and page layouts** to different users, based on their profile.

In simple terms:

👉 Record Types let you show **different versions of the same object** for different needs.

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◆ **Why do we use Record Types?**

- To support **different sales processes** (for Opportunities).
- To support **different business processes** (for Cases, Leads, Custom Objects).
- To show **different picklist values** to different users.
- To assign **different page layouts** to different profiles.

### Key Points

- **Profiles control access** → If a user's profile is not assigned a record type, they cannot create records of that type.
- **Default Record Type** → You can set a default for each profile.
- **One Object = Many Record Types** → You can create multiple record types per object.

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## 16. What is a Validation Rule in Salesforce?

A **Validation Rule** in Salesforce verifies that the data entered in a record meets certain criteria **before the record can be saved**.

👉 If the data doesn't meet the condition, Salesforce shows an **error message** and prevents the record from being saved.

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### ♦ Why do we use Validation Rules?

- To **maintain data quality**.
- To **prevent incorrect or incomplete data** from being saved.
- To **enforce business rules** consistently.

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## 17. Salesforce object Limitation

### ♦ 1. Custom Objects Limit

- **Essentials** → Up to **10** custom objects.
- **Professional** → Up to **50** custom objects.
- **Enterprise** → Up to **200** custom objects.
- **Unlimited / Performance** → Up to **2,000** custom objects.
- **Developer Edition** → Up to **400** custom objects.

(Additional objects can be purchased with Salesforce add-ons.)

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### ♦ 2. Fields per Object

- Each object (standard or custom) can have:
  - **Custom fields** → Up to **500 fields per object** (varies slightly by edition).

- Includes formula fields, roll-up summaries, and relationship fields.

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### ◆ 3. Relationships per Object

- Each object can have:
  - Up to **40 relationship fields** (25 lookups + 2 master-detail + 1 hierarchy on User object).
  - **Master-Detail relationships** → Max **2** per object (except junction objects which can have 2).
  - **Lookup relationships** → Up to **40 per object**.

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### ◆ 4. Record Types per Object

- Each object can have up to **200 Record Types**.

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### ◆ 5. Page Layouts per Object

- Each object can have up to **500 Page Layouts**.

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### ◆ 6. Validation Rules per Object

- Max **500 validation rules** per object.

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### ◆ 7. Other Important Limits

- **Formula Field** → Maximum of **3,900 characters** compiled size (and 5,000 for cross-object).
  - **Picklist Values** → Standard picklist can have up to **1,000 values**.
  - **Roll-up Summary Fields** → Max **25 per object**.
  - **Active Lookup Filters** → Max **5 per object**.
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## 18. What are Standard Objects and Custom Objects in Salesforce?

- **Standard Objects** → Built-in objects provided by Salesforce (e.g., Account, Contact, Opportunity, Case, Lead).
- **Custom Objects** → Objects created by admins/developers to store company-specific data (e.g., Student\_\_c, Invoice\_\_c).

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## 19. What are Tabs in Salesforce?

Tabs allow users to **navigate to an object's records** easily. For example, an "Accounts" tab lets you see all Account records. Tabs can be for **Standard Objects, Custom Objects, Visualforce pages, Lightning pages, or Web tabs**.

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## 20. What are Apps in Salesforce? What types of Apps can you create?

An **App** is a collection of objects, tabs, and other items that work together.

Types:

- **Standard Apps** (Sales, Service, Marketing).
- **Custom Apps** (built by admins/developers).
- **Console Apps** (optimized for service agents).

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## 21. Difference between Salesforce Classic and Lightning Experience.

- **Classic** → Old UI, limited features.
- **Lightning** → Modern UI, mobile-first, supports **Lightning Components, Lightning App Builder, Kanban Views, Path, Enhanced Reports**.

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## 22. What are Roll-Up Summary Fields? Limitations of Roll-up Summary Fields?

Roll-Up Summary fields calculate values (SUM, MIN, MAX, COUNT) from **child records** in a Master-Detail relationship.

Limitations:

- Only available on **Master objects**.
- Cannot be created on Lookup relationships.
- Max **25 per object**.

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## 23. What are Junction Objects? Why do we use them?

A **Junction Object** is a custom object with two **Master-Detail relationships**, used to create **many-to-many relationships**.

Example: Student\_\_c ↔ Course\_\_c → Enrollment\_\_c (Junction).

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## 24. What are External IDs in Salesforce?

- Fields that uniquely identify a record from an external system.
- Used during **data migration and integration**.
- Supported field types: Text, Number, Email.



## Matching Rule and Duplicate Rule

**Matching Rule:** Defines the criteria Salesforce uses to identify duplicate records (for example, comparing Email or Phone fields).

**Duplicate Rule:** Uses the matching rule(s) to determine what to do when a duplicate is found (Allow, Block, or Report).

### Use Case :

A sales rep tries to create a new Lead with the same email as an existing Contact.

Matching Rule → checks if the Email matches.

Duplicate Rule → blocks the save or allows it but shows a warning.

### Advantages

Prevents duplicate data entry.

Improves data quality.

Can be applied to standard and custom objects.

### Limitations

Maximum 5 active duplicate rules per object.

Matching is limited to the fields and logic Salesforce provides.

Cannot always detect “fuzzy” duplicates (like "Jon" vs "John").

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## Global Actions

### Explanation

Global actions let users create records, log calls, or send emails from anywhere in Salesforce (home page, Chatter feed, record page).

### Use Case

A sales rep can log a call or create a new Lead from the global action menu without navigating to the Lead tab.

### Advantages

Saves time by allowing quick access.

Provides consistent functionality across objects.

Useful in Salesforce mobile.



### **Limitations**

Limited customization compared to object-specific actions.

Cannot pre-populate values with object context (since it's "global").

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## **Global Picklists**

### **Explanation**

Global picklists (Picklist Value Sets) let admins reuse the same picklist values across multiple objects.

### **Use Case**

Status field across Leads, Opportunities, and a custom object can all use the same global picklist "Status" values (e.g., New, In Progress, Closed).

### **Advantages**

Centralized management (update values once, reflected everywhere).

Enforces consistency across objects.

Helps avoid data integrity issues.

### **Limitations**

Once deployed, global picklists are restricted (cannot add values directly on the field, must update in value set).

Cannot be converted back to a local picklist easily.

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## **Email Templates and Email Alerts**

### **Explanation**

Email Templates: Predefined email content (Text, HTML, Custom, Lightning templates).

Email Alerts: Workflow/Process Builder/Flow action to send emails using a template.

### **Use Case**

When a Case is escalated, an email alert sends a predefined template to the support manager.

### **Advantages**

Saves time with reusable templates.

Ensures consistent communication.

Can merge Salesforce fields dynamically (like {!Contact.Name}).

### **Limitations**

Classic and Lightning templates behave differently (Lightning templates not usable everywhere yet).

Limited design flexibility compared to external tools.

Mass emails have daily sending limits.

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## **Buttons, Links, and Actions**

### **Explanation**

**Buttons/Links:** Add custom navigation or functionality (URL, JavaScript, Visualforce, LWC).

**Actions:** Predefined record actions (Create, Update, Log a Call, Custom Lightning Component).

### **Use Case**

A custom button opens Google Maps with the Account address.

A quick action creates a new Opportunity from an Account record.

### **Advantages**

Enhances user productivity.

Allows integration with external apps (via URL or JavaScript).

Custom actions can simplify record creation.

### **Limitations**

JavaScript buttons are not supported in Lightning (need Lightning Components instead).

Too many buttons/actions can clutter the UI.

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## **What is the difference between 15 digit and 18 digit id in Salesforce?**

- 15 digit case-sensitive version
- 15 digit ID is what will be received from within the user interface
- 18 digit case-insensitive and best used for manipulating data outside Salesforce

- 18 digit id is referenced through in the API and other 3rd party tools.

## What are Governor Limits?

Since we are working in multitenant environment Salesforce is enforcing the limits for all the functionalities.

- How many no of fields for an object: 500
- No of master detail relationships for an object: 2
- No of lookup relationship for an object: 40 • No of rollup summary fields: 25
- How many external ids we can enable for an object: 7 (increased from 3 to 7 in Winter-15)
- Total number of records retrieved by a SOQL query – 50,000
- Total number of SOQL queries issued – 100 (Synchronous) 200 (Asynchronous)
- Total number of DML statements issued – 150
- Total number of callouts (HTTP requests or Web services calls) in a transaction – 100
- Maximum CPU time on the Salesforce servers – 10,000ms (Synchronous) 60,000ms

## What is an External ID in Salesforce ?

An external ID is a custom field that has the External ID attribute, meaning that it contains unique record identifiers from a system outside of Salesforce.

What is the maximum number of external IDs an object may have in Salesforce?

We can designate up to 25 External ID fields per object.

For which data type we can enable external id? text, number, auto number, email. --->> Use of External ID in Salesforce:

- It becomes searchable in the sidebar search.
- Salesforce record Id's are not required to load data.
- It can be used with UPSERT DML operation to integrate with other systems.
- The Data Import Wizard detects existing records in Salesforce with external IDs that match those values in the import file.

**Example:** When we update or upsert any record in Salesforce we do it by using Record Id. But if we want to update or upsert the record from some External System, we won't be having Record Id in the External system. So, we choose one field as 'External ID' in the Salesforce and based on that we can upsert the records from External System.

## What is a Lead?

URL from web-to-lead option. The generated HTML code can be hosted in any of the website. Upon entering the information in those fields and clicking on submit button that information will be saved. Leads are people who are interested in your product and service. Converting leads to loyal customers will provide success within a business.

## What is web-to-lead?

The process of using a website form to capture visitor information and store that information as a new lead in Salesforce. ---->> On lead object we can generate the HTML code by selecting lead fields and by mentioning return into lead object of the Salesforce. ---->>

## How many web-to-lead requests can you have in Salesforce?

The daily limit for Web-to-Lead requests is 500 per day. If your organization exceeds its daily Web-to-Lead limit, the Default Lead Creator receives an email containing the additional lead information.

### 1. Case in Salesforce

A **Case** is a **record that represents a customer's issue, question, or feedback** in Salesforce Service Cloud.

It helps support teams **track, manage, and resolve** customer problems.

Cases can be created manually by support agents or automatically (via email, web, phone, chat, etc.).

**Example:** A customer reports "Product not working after update." → A Case is logged with details like customer info, product, issue, and priority.

### Key Features

Case assignment rules → route to the right team/agent.

Escalation rules → ensure SLA timelines are met.

Case lifecycle → Open → In Progress → Resolved → Closed.

Integration with Knowledge Base → agents can suggest solutions quickly.

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### 2. Web-to-Case

**Web-to-Case** automatically creates a Case when a customer submits a form on your website.

#### How it works

You create a **form in Salesforce** and embed it on your website.

When customers submit the form → Salesforce creates a Case.

#### Example

On a company's support page, a customer fills a form:

Name: John

Email: john@test.com

Issue: "Login not working"

→ Salesforce automatically creates a Case with these details.

### **Advantages**

Free and built-in (no extra license).

Captures cases directly from website visitors.

Auto-response rules can send acknowledgment emails.

### **Limitations**

Cannot handle file attachments.

Limited to **5,000 cases/day**.

Exposes Org ID in the form code (may need security considerations).

---

## 3. Email-to-Case

**Email-to-Case** automatically creates a Case when a customer sends an email to a support address (like support@company.com).

### **How it works**

Salesforce monitors a support email address.

When a customer sends an email → Salesforce creates a Case and attaches the email content.

Replies from agents can also update the Case automatically.

### **Two Options**

**On-Demand Email-to-Case** → Salesforce servers handle emails.

Limit: 25 MB per email.

Simpler, no local setup.

**Email-to-Case (Agent)** → Requires an agent installed on a local server.

Can handle attachments > 25 MB.

Requires server maintenance.

### **Advantages**

Customers can just send an email (easy adoption).

Supports email attachments (like screenshots, documents).

Auto-response and workflow rules for faster handling.

### **Limitations**

Storage usage increases due to email attachments.

On-demand has a 25 MB email size limit.

Requires proper email-to-case routing and monitoring setup.

## Difference between custom setting and custom metadata :

### Custom Settings

Custom Settings are like **custom objects** that store data accessible across your org, but without needing SOQL queries (they're cached).

#### Types of Custom Settings

##### 1. List Custom Settings

- Acts like a custom object with rows (records).
- All users/org see the same data.
- Example: Store a list of **Tax Rates** or **Country Codes**.

##### 2. Hierarchy Custom Settings

- Data can be defined at **org, profile, or user level**.
- System checks in order: User → Profile → Org.
- Example: Different discount limits for **Sales Reps, Managers, and Admins**.

#### Use Cases

- Store org-wide configurations like support phone numbers.
- Control feature visibility (e.g., "Enable New Feature = True").
- Provide default values based on user profile (via Hierarchy).

#### Limitations

- Data is **not deployable** (you must recreate it in each org).
  - Limited storage (10 MB for List, 300 KB for Hierarchy).
  - Best for simple configs, not complex logic.
- 

## 2. Custom Metadata Types (CMDT)

Custom Metadata is similar to Custom Settings but is **deployable and packageable**.

#### Features

- Define a metadata type (like a custom object).
- Create records for it (like rows of data).
- Both the **type** and the **records** can be deployed via Change Sets, Metadata API, or packages.

#### Use Cases

- Store **business rules** (e.g., “Opportunity Stage to Probability Mapping”).
- Integration endpoints (like API URLs for different environments: Dev, QA, Prod).
- Feature toggles (Enable/Disable settings across orgs).
- Reusable framework data (e.g., error messages, validation rules).

### Advantages

- Deployable → consistent across sandboxes and production.
- Supports relationships with custom objects.
- Secure, doesn’t count against data storage.

| Aspect                       | Custom Settings                                                                                                          | Custom Metadata Types (CMDT)                                                                                             |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Definition</b>            | Special kind of custom object used to store org-wide or user-specific configuration data.                                | Custom objects that store configuration records as <b>metadata</b> instead of data.                                      |
| <b>Types Available</b>       | - <b>List Custom Setting</b> (org-wide data)- <b>Hierarchy Custom Setting</b> (user/profile/org-level data)              | Only one type, but you can create <b>Custom Metadata Types</b> and define multiple fields & relationships.               |
| <b>Deployment</b>            | Cannot deploy records (must recreate manually in each environment). Only the definition (object/fields) can be deployed. | Both <b>type (structure)</b> and <b>records (data)</b> are <b>deployable</b> via Change Sets, Metadata API, or Packages. |
| <b>Data Storage</b>          | Stored in <b>application cache</b> (fast access, doesn’t count against data storage).                                    | Stored as <b>metadata</b> , not data. Doesn’t count against data storage.                                                |
| <b>Access Method</b>         | Access via \$Setup or SOQL.<br>Example: CustomSettingName__c.getInstance(UserInfo.getUserId())                           | Access via SOQL.<br>Example: SELECT MasterLabel, Field__c FROM CustomMetadataType__mdt                                   |
| <b>Visibility / Security</b> | Data can be made public or private (depending on type). Can be restricted per profile/user (Hierarchy).                  | Always public and accessible in Apex. Cannot set profile/user-specific visibility.                                       |
| <b>Configuration Level</b>   | Supports org, profile, and user levels (Hierarchy).                                                                      | Same for all users/org (no                                                                                               |

| Aspect                      | Custom Settings                                                                                                                      | Custom Metadata Types (CMDT)                                                                                                                                                |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             |                                                                                                                                      | user/profile-specific variation).                                                                                                                                           |
| <b>Typical Use Cases</b>    | - User/profile-based limits (e.g., discount %)- Store static reference data (tax rates, phone numbers)- Feature enable/disable flags | - Environment-specific configs (API URLs)- Business rules (stage → probability mapping)- Error messages or validation rules- Configurations that need to be <b>deployed</b> |
| <b>SOQL Governor Limits</b> | Doesn't consume SOQL queries (cached).                                                                                               | Consumes SOQL queries (but usually small impact since records are few).                                                                                                     |
| <b>Record Count Limits</b>  | - List Custom Setting: 10 MB- Hierarchy Custom Setting: 300 KB                                                                       | No specific limit on records (but intended for config, not large datasets).                                                                                                 |

## Custom Label

Custom Labels are used to **store text values** that can be referenced in Apex, LWC, VF pages, validation rules, and formulas.

They support **multilingual translations**, so they're great for apps that need to work in multiple languages.

### Key Features

- Store up to **255 characters** (per label).
- Can be referenced in Apex (`System.Label.LabelName`), LWC (`$Label.c.LabelName`), and Visualforce (`{!$Label.LabelName}`).
- Can be translated using **Translation Workbench**.

### Use Cases

#### 1. Error Messages

- Instead of hardcoding error text in Apex, use labels.

Example: "You are not allowed to apply more than 10% discount"  
→ stored in a label.

If business changes the limit to 15%, just update the label, not the code.

#### 2. Multi-Language Support

- If your org runs in **English + French**, you can have one label:
  - English: "Submit"



- French: “Soumettre”
    - Salesforce will show the right translation automatically.
  - 3. **Reusable Static Text**
    - API endpoints, button names, UI text, company tagline, etc.
- 

## 2. Custom Permission

Custom Permissions allow you to **control access to specific functionality** in Salesforce **without relying only on profiles or permission sets**.

### Key Features

- Defined in **Setup → Custom Permissions**.
- Added to **Profiles** or **Permission Sets**.
- Checked in Apex, Validation Rules, Flows, or even LWC.
- Helps avoid creating multiple profiles/permission sets just for one feature.

### Use Cases

1. **Feature Access Control**
  - Example: Only some users should see a “Generate Report” button.
  - Instead of checking the profile/role, you assign a custom permission “CanGenerateReport”.
  - Button is visible only if the user has that custom permission.
2. **Validation Rule Bypass**
  - Example: Discount cannot exceed 10%.
  - Validation Rule:
  - ```
AND(
  Discount__c > 0.10,
  NOT($Permission.Bypass_Discount_Validation)
)
```
  - Now, Admins with the **Bypass\_Discount\_Validation** custom permission can save records with higher discounts.

## Data Import Wizard:

A **web-based tool inside Salesforce** (Setup → Data Import Wizard).

- Best for **simple, smaller data loads**.
- Guided step-by-step interface.
- Designed for **Admins & Business Users** (non-developers).

### Key Features

- Import up to **50,000 records** at a time.

- Supports standard objects (Accounts, Contacts, Leads, Solutions, Campaign Members, Person Accounts).
- Supports **all custom objects**.
- Can **insert, update, upsert** records (no delete/hard delete).
- Field mapping assistance (auto-match fields).

### Use Cases

- Importing a **new list of leads** from an event.
- Uploading **product catalog** into a custom object.
- Updating customer contact details (small volume).

### Advantages

- Simple, no installation needed.
- Friendly UI with guided steps.
- Prevents duplicates with matching rules.

### Limitations

- Max 50k records.
- Cannot delete/hard delete records.
- Limited scheduling (must run manually).

---

## Data Loader

A **client application (desktop tool)** used for **large, complex data operations**.

- Best for **Admins & Developers** handling high-volume data.

### Key Features

- Handles up to **5 million records** at a time.
- Supports **Insert, Update, Upsert, Delete, Hard Delete, Export**.
- Can be run via **command line** (scheduling with scripts).
- Works with **all standard & custom objects**.

### Use Cases

- Migrating millions of Accounts & Contacts from legacy systems.
- Bulk updating Opportunities with new values.
- Deleting old test records from sandbox.
- Exporting records for backup.

### Advantages

- Handles very large volumes.

- Supports **delete & hard delete**.
- Can automate/schedule jobs with CLI.

### Limitations

- Requires installation.
- Needs more technical knowledge.
- Manual field mapping for first use.

## Key Differences: Data Import Wizard vs Data Loader

| Aspect                       | Data Import Wizard                                                                                                               | Data Loader                                                               |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>Tool Type</b>             | Web-based tool inside Salesforce (no install needed).                                                                            | Client application (must be installed on desktop).                        |
| <b>Record Limit</b>          | Up to <b>50,000 records</b> per job.                                                                                             | Up to <b>5 million records</b> per job.                                   |
| <b>Supported Objects</b>     | Limited Standard Objects (Leads, Accounts, Contacts, Solutions, Campaign Members, Person Accounts) + <b>All Custom Objects</b> . | <b>All Standard and Custom Objects</b> .                                  |
| <b>Operations Supported</b>  | Insert, Update, Upsert.                                                                                                          | Insert, Update, Upsert, <b>Delete, Hard Delete, Export</b> .              |
| <b>Ease of Use</b>           | Simple, guided UI (non-technical users).                                                                                         | Requires technical knowledge, best for Admins/Devs.                       |
| <b>Duplicates Handling</b>   | Supports duplicate prevention via Matching Rules.                                                                                | No built-in duplicate prevention (must handle manually).                  |
| <b>Automation/Scheduling</b> | Not possible (manual only).                                                                                                      | Supports <b>command line automation &amp; scheduling</b> .                |
| <b>Data Volume Handling</b>  | Best for <b>small to medium data loads</b> .                                                                                     | Best for <b>large data migrations &amp; bulk operations</b> .             |
| <b>Installation</b>          | Not required (runs in browser).                                                                                                  | Required (Windows/Mac installer).                                         |
| <b>Typical Use Case</b>      | Import a small list of Leads from an event, update Contact details.                                                              | Migrate millions of records, mass delete old data, backup/export records. |

### Dynamic form :

Dynamic Forms let you **configure record detail fields and sections directly in the Lightning App Builder**, instead of being stuck with the standard page layout.

In simple words:

You can make record pages more **flexible, modern, and user-friendly** by deciding **which fields/sections appear, and for whom, under what conditions**.

---

## Key Features of Dynamic Forms

- Available for **custom objects** and some **standard objects** (like Account, Contact, Opportunity, Case).
  - Fields and sections are treated as **components** in the Lightning App Builder.
  - You can apply **visibility rules** (show/hide fields/sections based on criteria like user profile, record field value, or device type).
  - No need to maintain multiple page layouts for different profiles → reduce admin overhead.
  - Provides **Field Section** component where you can drag & drop fields.
- 

## How it Works (Step by Step)

1. Go to **Lightning App Builder** for a record page.
  2. Replace the old "**Record Details**" component with "**Field Section**".
  3. Drag & drop fields from the sidebar into sections.
  4. Apply **visibility rules** (example: only show "Discount %" field if "User Profile = Sales Manager").
  5. Save & Activate the page.
- 

## Use Cases

1. **Conditional Field Visibility**
    - Show "Reason for Discount" field only if Discount > 10%.
  2. **Profile/User-Specific Layouts**
    - For Support Agents → Show fields related to Customer Issues.
    - For Managers → Show approval-related fields only.
  3. **Simplify Data Entry**
    - Hide irrelevant sections (like "Billing Details") for users in roles that don't need them.
  4. **Mobile Optimization**
    - Show fewer fields on mobile vs full details on desktop.
- 

## Approval process:

An **Approval Process** automates how **records are approved in Salesforce**. It defines a **sequence of steps** that a record goes through until it is **approved, rejected, or recalled**.

It's typically used for business scenarios where **managerial or multi-level approval** is required before taking action.

---

## Key Components of an Approval Process

1. **Entry Criteria** – Which records qualify for approval (e.g., Opportunity Discount > 20%).
  2. **Approvers** – Who will approve (user, role, queue, or related user like Manager).
  3. **Steps** – Each level of approval (single or multi-step).
  4. **Actions** – What happens at each stage:
    - **Initial Submission Actions** → when record is submitted.
    - **Approval Actions** → when record is approved at a step.
    - **Rejection Actions** → when record is rejected.
    - **Final Approval/Final Rejection Actions** → when process ends.
    - **Recall Actions** → when submitter recalls request.
- 

## How it Works (Flow)

1. A record meets the entry criteria.
2. User submits the record for approval.
3. Record locks (cannot be edited unless unlocked).
4. Approval request is routed to the designated approver(s).
5. Approver can **Approve** or **Reject**.
6. Depending on the decision, actions are executed.

## What is a Report in Salesforce:

A **Report** in Salesforce is a **list of records that meet certain criteria**, displayed in a structured format.

- It helps analyze data, track performance, and make business decisions.
  - Reports are stored in **folders**, which control access (who can view, edit, or manage).
- 

## Key Features of Reports

- Built on **Report Types** (decide which objects and fields are available).

- Can include **filters** to show only relevant records.
  - Data is displayed in **rows and columns** (like Excel).
  - Can add **charts** for visualization.
  - Reports can be scheduled and sent by email.
- 

## Types of Reports in Salesforce

### 1. Tabular Report

- Simple list of records with fields in rows & columns.
  - No groupings, just plain data.
  - Example: List of all Contacts with Phone & Email.
  - Best for **exporting data** or **mailing lists**.
- 

### 2. Summary Report

- Allows grouping of rows based on a field.
  - Can show subtotals.
  - Example: Opportunities grouped by **Stage** → shows revenue per stage.
  - Best for **subtotals and grouped analysis**.
- 

### 3. Matrix Report

- Like Summary, but allows grouping by **rows and columns**.
  - Shows data in a grid format.
  - Example: Opportunities grouped by **Stage (rows)** and **Owner (columns)** → shows revenue by stage for each rep.
  - Best for **comparisons across multiple dimensions**.
- 

### 4. Joined Report

- Combines multiple report blocks (based on different report types).
  - Allows viewing data from different objects together.
  - Example: Opportunities + Cases in one report to see revenue vs customer support.
  - Best for **multi-object analysis**.
- 

## Dashboard

A **Dashboard** in Salesforce is a **visual representation of one or more reports**.

- It helps users quickly understand **key metrics** and business performance.
- Each component in a dashboard is based on a **report**.

- Think of it like: *Reports = data, Dashboard = visual story of that data.*
- 

## Key Features of Dashboards

- Built from **Reports** (must exist first).
  - Can have multiple **components** (charts, tables, gauges, metrics).
  - **Filters** allow users to adjust the view (e.g., by region, product).
  - **Dynamic Dashboards** show data according to the logged-in user's security settings.
  - Dashboards are stored in **folders** (like reports).
- 

## Types of Dashboard Components

1. **Chart** – Bar, Line, Pie, Donut, Funnel → visualize trends.
  2. **Gauge** – Shows a single value within a range (like speedometer).
  3. **Metric** – Displays a single key value (like total revenue).
  4. **Table** – Shows detailed data from reports.
  5. **Lightning Table/Chart** – More interactive in Lightning Experience.
- 

## Types of Dashboards

1. **Standard Dashboard**
  - Same view for all users.
  - Example: Sales Performance dashboard visible to Sales team.
2. **Dynamic Dashboard**
  - Shows data **as per logged-in user's security and sharing settings**.
  - Example: Each Sales Rep sees only their own pipeline.
  - Useful for avoiding multiple dashboards for different users.